

Measurements of Waves in Landfast Ice Using Inertial Motion Units

Jean Rabault, Graig Sutherland, Brian Ward, Kai H. Christensen, Trygve Halsne, and Atle Jensen

Abstract—Inertial motion units (IMUs) are used to perform measurements of waves in water covered by landfast ice close to the ice edge in Svalbard. The effective noise level of the instruments is assessed in controlled wave tank experiments. A set of measurements collected in Tempelfjorden, Svalbard in March 2015 is presented, and the ability of the sensors to operate in the field is validated. Several characteristics of the recorded signals, including correlation between the different sensors, are analyzed. Horizontal and vertical motions are of the same order of magnitude. A clear transition in the signal properties is observed in relation with changes in incoming wave field and the development of cracks in the ice layer. We show that complex physics takes place when waves propagate in landfast ice and that the use of times series containing information on the full three-dimensional linear acceleration, rather than spectra, is required to capture the underlying phenomena.

Index Terms—Inertial motion units (IMUs), polar environment, wave propagation, waves in ice.

I. INTRODUCTION

THE impact of sea-ice cover on surface wave propagation has been studied for over five decades, both theoretically and in the field [1], [2]. Understanding the interaction between surface waves and sea-ice covers is a topic of interest for a variety of purposes such as the formulation of ocean models for climate, weather, and sea state prediction [3], estimation of ice thickness [4], and the analysis of pollution dispersion in the Arctic environment [5], [6]. Unfortunately, our knowledge of the underlying phenomena is still partial as pointed out by the difficulties encountered when trying, for example, to model climate and ice extent evolution [7]. Detailed theoretical analyses of wave damping by sea ice in idealized situations are available in the literature [8]–[11]. Data from laboratory experiments, although introducing some limitations, are also available [12]. However, the experimental data available for validating the theoretical results and studying the processes involved in water wave damping by ice in real conditions are still limited. Although some studies tried to compare field data

with experiments [13], more measurements must be done in the field to generate enough data that will allow the gap to be closed between theory, laboratory experiments, and ocean dynamics.

Historically, measurements of wave propagation in sea ice have been done using a variety of instruments that have evolved with the development of better electronics and sensors. Early work involved the use of strain meters and accelerometers for analyzing large floe dynamics [14], [15]. In the late 1980s and early 1990s, studies using accelerometers for capturing wave motion in ice were presented [16], [17]. The amount of data available at that time was still limited due to the cost of such devices, which prevented large deployments on the field. More recently, due to developments in microelectronics, ranging from better sensors to more powerful embedded computers and improved communication, new possibilities for performing direct measurements of waves in ice opened. GPS positioning, high-accuracy tilt-meter information, and the possibilities offered by the Iridium satellite network made it possible to collect wave data from polar regions in an automated way [18]. Data on wave damping and drifting of ice floes in the Antarctic region have been gathered using such methods [19]–[21]. Attenuation of waves in the marginal ice zone has also been reported using similar techniques [22]. However, the data obtained in those studies are limited in several ways. Only one component of the acceleration or the angular position was recorded, and the corresponding data were retrieved through transmitting Fourier spectra. This last point implies a reduction of the information content in the data, effectively filtering out, for example, wave modulation and grouping that have proved several times to be a possibly important feature for ice breakup [22]–[24].

The evolution of available consumer electronics implies that it is now possible to record high-accuracy full dynamic information, i.e., three-axis linear accelerations, using a single affordable inertial motion unit (IMU) [25]. In this paper, we report a detailed assessment in the laboratory and use on the ice of such engineering electronic components for measuring waves in landfast ice. The ability of the IMU to measure wave motion is demonstrated in controlled wave tank experiments. A series of measurements in Tempelfjorden, Svalbard, during which waves were successfully detected in landfast ice, is reported and analyzed. Horizontal accelerations are found to be the same order of magnitude as vertical accelerations. This is surprising, as horizontal motion in a solid ice cover had not been previously reported, and the authors are unsure about the origin of this motion. A transition between wind-generated waves and swell is also detected, and it was proven in another article about the same data set that the appearance of cracks in the ice cover around the time when maximum significant wave height is attained leads to changes in wave propagation [26]. Significant changes in the level of reflected wave energy

Manuscript received January 29, 2016; revised May 12, 2016; accepted June 7, 2016. Date of publication July 14, 2016; date of current version September 27, 2016. This work was supported by The Research Council of Norway, PETROMAKS 2 Program; Grant 233901, "Experiments on Waves in Oil and Ice."

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Digital Object Identifier 10.1109/TGRS.2016.2584182

from the ice cover are observed in relation to this evolution in the incoming waves and ice cover. Changes in the wave signal and wave envelope correlations between the different IMUs are also observed during this transition, which can be explained by analyzing the dispersion relation for waves in ice.

The outline of this paper is as follows. First, the IMU model is assessed in the laboratory. Following this, information on the IMUs' deployment conditions, including wave information from numerical models, is presented. The signal recorded is then compared with the wave model data, and emphasis is put on the detection of a change in the wave spectrum. Finally, the signals recorded by the different sensors are compared to analyze wave propagation.

II. LABORATORY ASSESSMENT OF THE IMU

The IMUs used in this work are thermally calibrated rugged VN-100 IMU systems, based on solid-state microelectromechanical system sensors and produced by Vectornav Co. They combine a three-axis accelerometer, a three-axis gyroscope, a three-axis magnetometer, a barometric pressure sensor, a thermometer, and a 32-bit processor. Thermal calibration allows an operating range of -40°C to 85°C . An output rate of 10 Hz was chosen for performing measurements. Data quality was checked during the processing by the 16-bit CRC16-CCITT algorithm. According to the technical documentation, the resolution obtained for acceleration is better than $5 \times 10^{-3} \text{ m/s}^2$ [27]. In this paper, we will only analyze the linear acceleration signals.

The capability of the VN-100 IMU to measure accelerations typical of waves in ice was assessed in wave tank experiments in the University of Oslo Hydrodynamics Laboratory. The wave tank is 24.6 m long and 0.5 m wide, and the water depth is 0.6 m [28], [29]. The wave generator consists of a vertical plate driven by a computer-controlled hydraulic piston. At the far end of the tank, an absorbing beach reduces the amplitude of reflected waves to less than 3% of the incoming amplitude. The motion of the wave maker during the measurements is chosen to be sinusoidal with constant amplitude to generate harmonic waves, and a brief transient amplitude following a tanh law is used at the beginning of the run to establish the waves. Each run lasted for 10 min, so that the waves reach a steady-state harmonic regime when measurements are performed.

The IMU was fixed on a 30-cm-long rectangular foam floater. The floater was prevented from drifting with the waves using a rope fixed 8 m ahead on the wave tank, so that the force on the floater was primarily horizontal and the vertical motion of the float was not affected. As a consequence, the floater motion is the superposition of a vertical displacement that follows water elevation and a pitch oscillation caused by wave slope. No twisting motion is observed. Data are sent to the logging PC using a radio frequency (RF) transmitter, to avoid any electrical wire adding rigidity in the system. The typical vertical acceleration associated with representative waves in ice of amplitude $a = 10 \text{ cm}$ and period $T = 2\pi/\omega = 6 - 10 \text{ s}$ is $A_z = a\omega^2$ and has a magnitude of 0.04 to 0.1 m/s^2 . Waves of period $T = 2 \text{ s}$ and amplitude 3 mm are generated in the wave tank, and the corresponding vertical accelerations generated have an amplitude of 0.03 m/s^2 . The discrete Fourier transform (DFT) of the signal obtained is computed using a 100-s time window and is presented together with a sample of the raw data in Fig. 1.

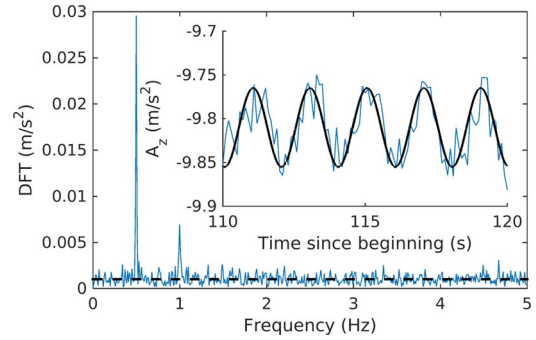


Fig. 1. Laboratory assessment of the IMUs. DFT of the wave vertical acceleration from a 100-s signal sample and a 10-s sample of the signal (A_z , inset). The float moves with monochromatic water waves generating vertical accelerations representative of values expected in the field. Blue: signal recorded by the IMU. Black: spectral noise level of 0.001 m/s^2 and theoretical sinusoidal wave (inset).

The main peak in the DFT of Fig. 1 is observed at a frequency of 0.5 Hz, which corresponds to the waves generated. The peak value is one order of magnitude over the noise level experimentally observed (black dashed line). The wave vertical acceleration amplitude obtained by the DFT at 0.5 Hz is 0.0296 m/s^2 , which corresponds to a relative error of around 1.3% relative to the waves generated. The secondary peaks that can be observed for frequencies of 1 and 1.5 Hz are higher-order harmonics that come from the intrinsic dynamics of the float and nonlinearities introduced by the wave generator. Those measurements prove the ability of the sensors to measure waves with typical vertical acceleration corresponding to what is expected in the field.

III. FIELD MEASUREMENTS

A. Instrument Deployment and Local Ice and Wave Conditions

The instruments were deployed in Tempelfjorden, Svalbard between March 25 and March 27, 2015 (see Fig. 2 for positioning in Tempelfjorden). The local bathymetry of the fjord is also presented in Fig. 2, together with the relative positioning of the IMUs and a picture of the ice cover taken from an adjacent mountain. A layer of thin fresh ice was present between the thick ice edge and the open water, the thickness of which was estimated to be around 20 cm, with a horizontal extent of about 500 m. The ice thickness further back from the ice edge ranged between 60 and 80 cm. The main data logger was located around 180 m from the thick ice edge.

The main data logger was a wide temperature Moxa UC8410 computer [30]. It recorded the acceleration measured by five VN-100 IMUs, as well as GPS information and was also used to write time stamps on the data collected. One of the IMUs was collocated with the data logger. Two IMUs were deployed east and west at distances of about 10 m from the data logger, using cables for power supply and data transmission. Two more were deployed West 50 m away from the logger and used Li-Ion batteries for the power supply and RF transmitters for communication with the Moxa. The IMUs were protected in plastic hard cases that were buried under the snow so that they were in direct contact with the ice. The orientation of the IMUs was chosen so that their x -axis would be recording acceleration along the direction of the fjord, whereas the y -axis is perpendicular to the direction of the fjord, and the z -axis points downward. The RF transmitters used are YS-C10U units from USConverters [31] running on a 5-V power supply. The

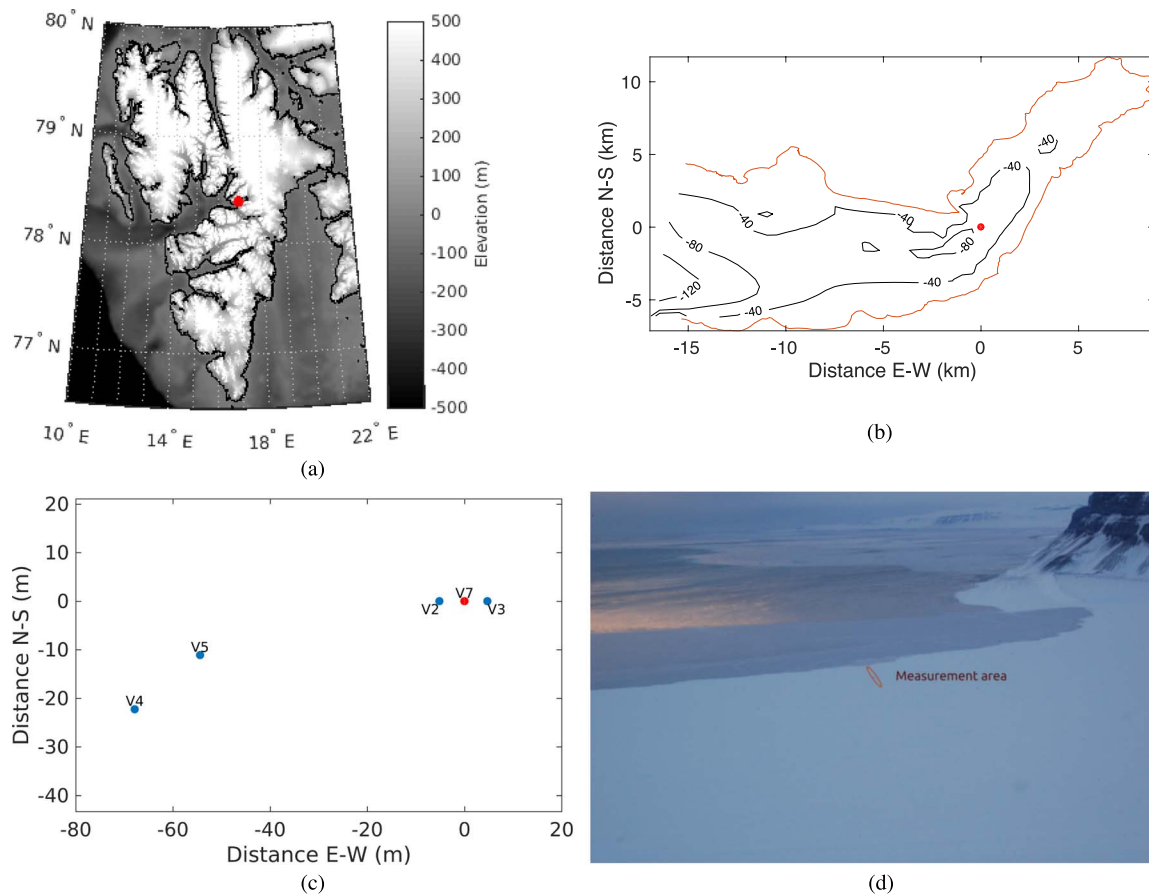


Fig. 2. (a) Location of the measurements area in Tempelfjorden, Svalbard. The positioning of the Moxa logger is indicated by the red dot. (b) Details about Tempelfjorden bathymetry. The local map is centered on the position of the Moxa data logger (red dot). (c) Relative positioning of the IMUs. The local map is centered on the position of the Moxa data logger (red dot). (d) Overview of the measurement area. The picture is taken from a mountain adjacent to the fjord. The location of the loggers is indicated by the red ellipsis.

RF modules emit at a frequency of 433 MHz with 12 different channels available. A pair of emitter/receiver modules works as a wireless serial data transmission line, of which the Bd rate can be up to 115 200 b/s (bits per second). In the present experiment, a Bd rate of 9600 b/s was enough to transmit the data. The temperature range for the transmitters is from -35°C to 75°C , and the emitting power is 50 mW. While this was sufficient to transmit over a couple hundred meters of line of sight on ice-free mainland, the effective transmission distance in the field was only up to 75 m. A likely explanation for the reduced range is the difficulty to get a reliable power supply and signal reflections due to ice and snow.

A Davis Vantage Pro2 Plus weather station was used to collect hourly records of air temperature as well as wind strength and direction. The orientation of the weather station is determined by a magnetometer, which may introduce some uncertainties at the latitude where the measurements were performed, but relative changes in wind direction can be recorded with an accuracy level of 3° .

The instruments were deployed during the day of March 25, 2015. The weather conditions changed from calm weather to storm conditions on March 26, and waves propagated into the fjord during several hours starting in late afternoon the same day. A team went to check the instrument function in the middle of the day and noted the appearance of cracks in the thick ice. The instruments were recovered on March 27,

and numerous cracks were observed all over the measurement area during recovery. The evolution of the wave field further offshore, obtained from the numerical wave prediction model WAM of the European Centre for Medium Range Weather Forecasts (ECMWF), is presented in Fig. 3. T_0 corresponds to the initial deployment of the instruments, i.e., 15:00 UTC on March 25, and will, in the following, be used as the reference time. As seen in Fig. 3, the wave field progressively gets aligned with the direction of the main fjord system, which makes the link between Tempelfjorden and the open sea and increases in strength up to around $T_0 + 30$ h before declining. Data from the ECMWF model also indicate that the wave peak period at the entrance of the fjords system (12.9°E , 78.10°N), obtained from the 1D spectrum peak period (PP1D), changes from around 6.5 s at T_0 to over 10 s around $T_0 + 45$ h. The evolution of significant wave height at the entrance to the fjord system is presented in Fig. 4. The wave height obtained in Tempelfjorden is significantly lower than outside of the fjord and is estimated in the range of 10 cm from the measurements performed, but the time evolution of the level in wave energy and peak frequency follows the same trends, as will be shown in the next section.

B. Wave Detection in the Ice

Due to the conditions imposed on the equipment and batteries, including temperatures below -20°C and an incoming

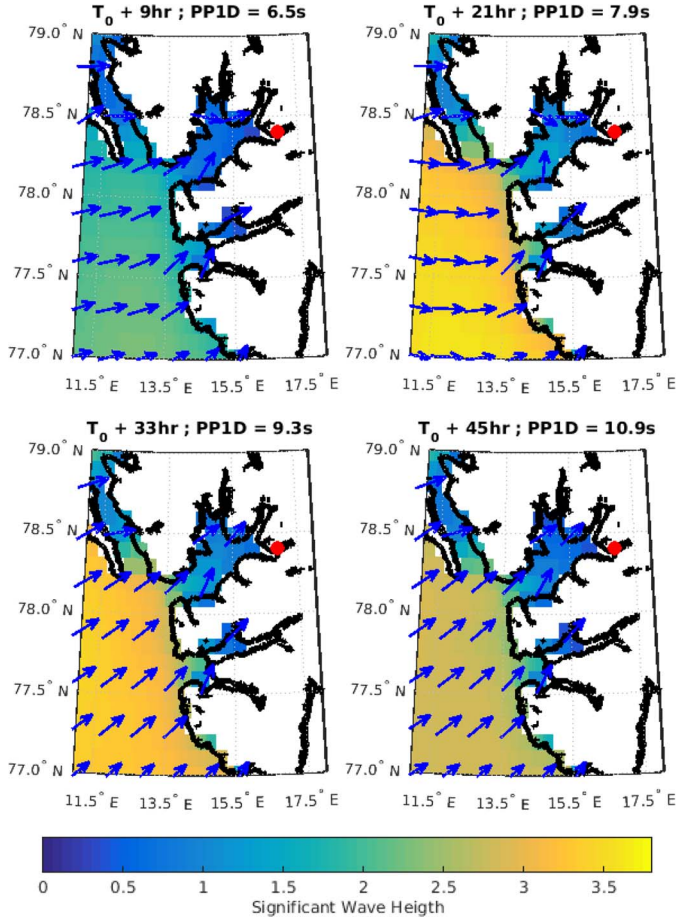


Fig. 3. Evolution of the wave system with time, obtained from the ECMWF model. Color indicates significant wave height, and arrows indicate the direction of propagation of the waves. The position of the sensors is indicated by the red dot. T_0 is the time of the beginning of the time series, i.e., 25/03/2015 at 15.00.00 UTC. PP1D is the peak period of the one-dimensional wave spectrum, at the position 12.9° E, 78.10° N.

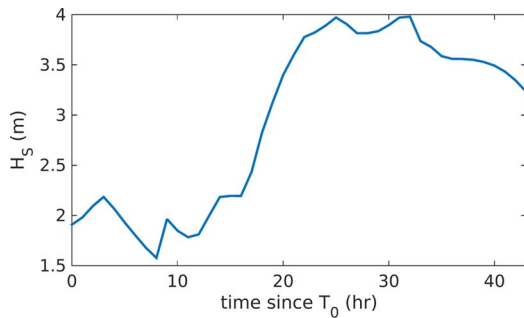


Fig. 4. Significant wave height obtained from the ECMWF model, at a position (77.9° N, 13.9° E) at the entrance of the fjord system where the measurements were performed.

storm, some of the sensors stopped working. The main points explaining those data losses were failure in the RF communication and low performance of the battery sets used in cold conditions. The GPS and weather station gave complete information during the two days, but part of the IMU data was lost. The internal IMU attached to the Moxa as well as two wireless IMUs and one wired IMU worked at least during part of the experiment. The data collected are summarized in Table I. The complete duration of the measurements is around 43 h. An

TABLE I
SUMMARY OF DATA COLLECTED

Sensor	Valid data time extend	Percentage bad data points
Weather station	$T_0 + 0 - 43$ hr	No bad data
V7 (with Moxa)	$T_0 + 0 - 43$ hr	<5%
V2 (wired 1)	$T_0 + 0 - 43$ hr	<5%
V3 (wired 2)	$T_0 + 0 - 21$ hr	<5%
V4 (wireless 1)	$T_0 + 0 - 35$ hr	>30%
V5 (wireless 2)	$T_0 + 24 - 39$ hr	<10%

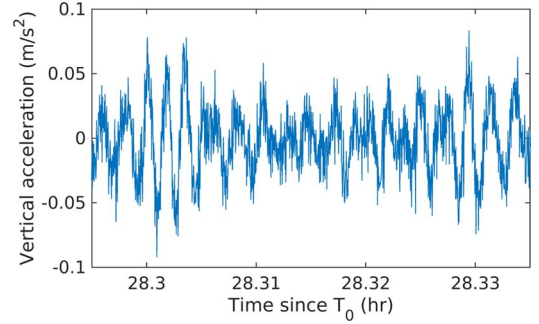


Fig. 5. Detail of raw data corresponding to the wave-induced vertical acceleration recorded during wave activity by the accelerometer V2.

interruption in the data is present around 24 h after the start and corresponds to the team checking the instruments and changing the power supply.

Clear wave motion was recorded, as evident from the raw data for vertical acceleration shown in Fig. 5. Following the method for analyzing waves in both ice-free and ice-covered water presented in [22], [32], and [33], data from the IMUs were further analyzed by means of Fourier transform and power spectral density (PSD) using the Welch method based on 30-min windows. The raw signal, from which corrupted data points are suppressed using checksum control by the CRC16-CCITT algorithm and interpolated from the neighboring valid points, is first bandpass filtered and, afterward, used for computing the PSD. Bandpass filtering is a fifth-order Butterworth filter, with cutoff frequencies of 0.025 and 0.5 Hz. Since the vertical acceleration is measured by the IMUs, the PSD associated to the surface elevation is calculated using the following relation (see [34]):

$$\text{PSD}[\eta] = \omega^{-4} \text{PSD}[\eta_{tt}] \quad (1)$$

where η is the surface elevation, ω indicates frequency in radians per second, and PSD is the power spectral density operator. This transform tends to amplify noise at low frequencies. Fig. 6 presents the comparison of the PSD spectra obtained from the vertical acceleration and surface elevation recorded by the accelerometer V2 before the arrival of waves in the fjord ($T_0 + 10$ h) and during part of the wave event ($T_0 + 27.75$ h). The ω^4 factor leads to increased noise in the low-frequency part of the spectrum, and the energy peak at high frequency is attenuated compared with low-frequency motion in the case when waves are detected. As a consequence of the high level of noise introduced at low frequency, the spectra presented will not always be corrected for the ω^4 factor. One should then keep in mind that the positions of the energy peaks stay unaffected, but that their relative heights are changed.

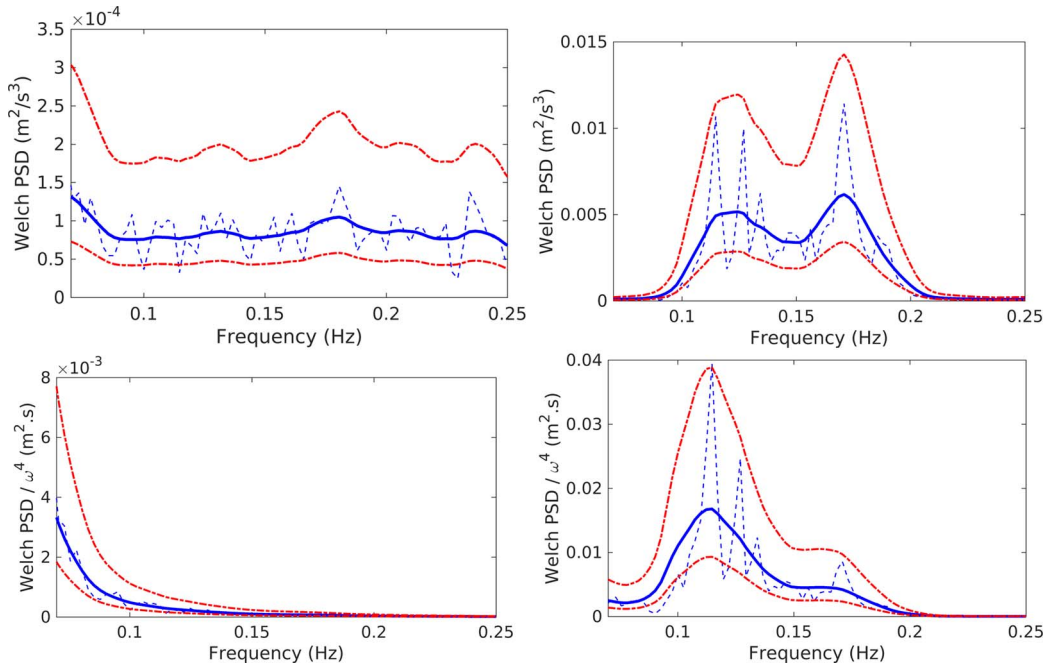


Fig. 6. PSD obtained from the Welch method without (left, $T_0 + 10$ h to $T_0 + 10.5$ h) and with (right, $T_0 + 27.75$ h to $T_0 + 28.25$ h) wave signal, from V7 time series. (Top) PSD of the vertical acceleration. (Bottom) PSD of surface elevation obtained by dividing the vertical acceleration spectra by ω^4 . Thin broken blue curves indicate the output of the Welch method, whereas thicker blue curves are smoothed results. Red curves indicate the limits of the 95% confidence intervals.

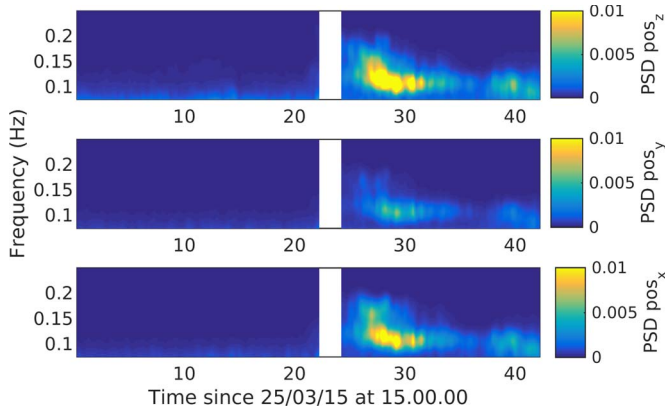


Fig. 7. PSD of the linear accelerations recorded by accelerometer V2 computed by the Welch method based on 30-min time windows with a 50% overlap and divided by ω^4 to obtain the spectrograms of the PSD for ice elevation, x -position and y -position.

Spectrograms are used for performing further analysis of the data. The raw signal, from which bad data points are suppressed using the checksum and interpolated, is divided in 30-min sequences with a 50% overlap. The PSD of η is computed on each data chunk using the Welch technique and (1). Finally, the obtained spectra are filtered using a sliding mean average of width 5 points to smooth the signal. Clear wave motion is observed (see Fig. 7). It is interesting to note that the PSD for the horizontal displacement along the fjord direction is of the same order of magnitude as the one for elevation. This implies that the ice moves not only up and down but also horizontally in the direction of wave propagation. This horizontal motion is puzzling to us as it is, to our knowledge, not mentioned in the literature. While the ice is landfast, the distance to the coast is

on the order of 2 km so that the relative deformation needed to explain for this motion is very small. In addition, cracks, ridges, and other irregularities in the ice cover could be a source of such horizontal motion. Although the wave motion is primarily along the fjord direction, there is also residual motion across the fjord direction. Several mechanisms may explain the remaining small motions recorded in the across fjord direction. Imperfections in the alignment of the IMUs along the direction of the fjord and the generation of nonplanar, diffracted, and reflected waves due to the topography can be the origin to the effects observed (see Fig. 2).

The spectrogram for η , obtained from V2 using (1), along with information on wind direction and strength recorded by the weather station, is shown again in Fig. 8 using the standard meteorological convention. A transition in the frequency of the signal recorded is observed around $T_0 + 27$ h in the elevation spectrogram. The wind gets aligned along the fjord at $T_0 + 21$ h, and the wind speed steadily increases to a maximum around $T_0 + 24$ h. The wind starts decreasing from $T_0 + 27$ h. This period of higher winds corresponds to the high-frequency peak observed in the spectrograms in Fig. 8. After 27 h, the wind velocity decreases. According to the ECMWF model (see Fig. 4 in Section III-A), the incoming swell starts building up around the same time, so that $T_0 + 27$ h is a transition between wind-generated waves and swell. After $T_0 + 27$ h, the spectral peak has shifted to a lower frequency. All those changes are in close agreement with the results from the ECMWF model.

The relation between horizontal and vertical accelerations is further analyzed by looking at the complex quantity, i.e.,

$$Y = A_x + iA_z \quad (2)$$

where A_x is the horizontal component and A_z the vertical component of the acceleration. The rotary spectrum $\text{PSD}[Y]$ is

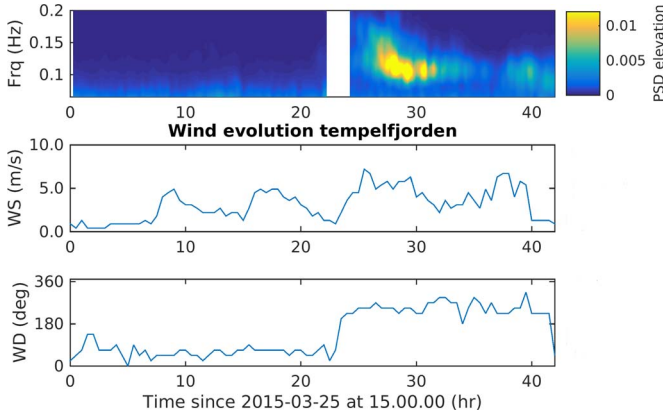


Fig. 8. Spectrogram of PSD for elevation (corrected by ω^4) obtained from V2, along with wind information from the weather station deployed at the measurement place.

computed [35], [36] using the same procedure previously presented for the IMU V2. V2 is chosen since the signal obtained is of high quality and separated from the main logger, which, together with the batteries, represents a heavy load on the ice. The double-sided spectra obtained for $T_0 + 26$ h and $T_0 + 29$ h are presented in Fig. 9. Since double-sided spectrograms are used, the division by ω^4 is not used to correct the shape of the spectra to avoid introducing a singularity around the zero frequency. One should therefore keep in mind that the PSD displayed magnify the importance of high-frequency content compared with low frequency. Most of the spectral energy density is in the positive-frequency domain, corresponding to the propagation of waves into the ice [36]. There is, however, spectral energy in the negative-frequency domain, corresponding to reflected waves propagating toward open water. It is interesting to note that the relative level of reflected energy is higher for high-frequency waves, in the early part of the wave activity. This corresponds to the fact that high-frequency waves are more affected by the ice cover than lower-frequency waves. One of the possible origins for the observed reflection is inhomogeneities in the ice sheet, such as cracks and pressure ridges. Theoretical studies [37]–[40] have predicted that defects in the ice sheet frequency-dependent reflection effects will increase with higher frequency, in agreement with our observation.

IV. WAVE PROPAGATION BETWEEN THE SENSORS

The motions recorded clearly indicate that waves come in packets (see Fig. 5). Detailed analysis of modulated wave groups can involve a variety of techniques [41]–[43]. Here, we obtain the shape of the wave envelope obtained from the Hilbert transform of the signal [44]. Obtaining accurate estimate envelope requires the signal to be low-pass filtered and smoothed, which is performed using the same bandpass filter as in Section III-B, followed by a sliding average filter with 10 points half width. Typical results are indicated in Fig. 10, demonstrating the presence of clearly defined wave packets with envelopes that are easily detected by the Hilbert transform.

Since the time series is available, it is possible to study the correlation between the different sensors. For this, the

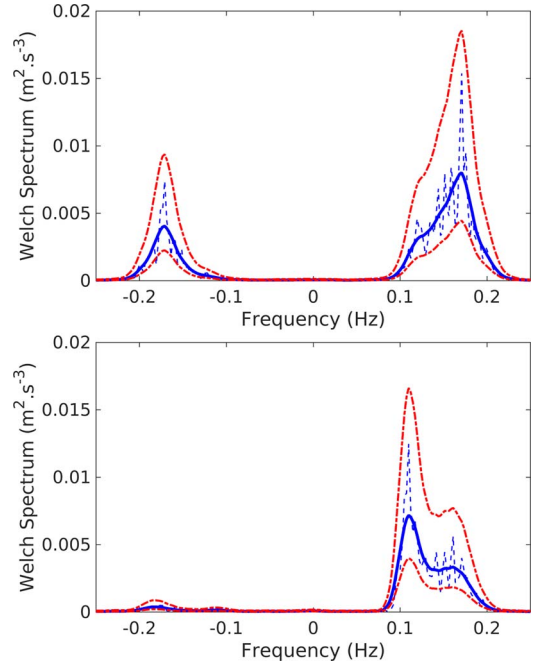


Fig. 9. Two-sided PSD, obtained by the Welch method with a 30-min time window, of complex quantity Y during the period of peak activity using the times series from V2. The correction by ω^4 is not included to avoid introducing noise at low frequency. (Top) Spectrum at $T_0 + 26$ h, when wind waves dominate. (Bottom) Spectrum at $T_0 + 29$ h, when lower-frequency swell dominates. Thin broken blue curves indicate the output of the Welch method, whereas thicker blue curves are smoothed results. Red curves indicate the limits of the 95% confidence intervals.

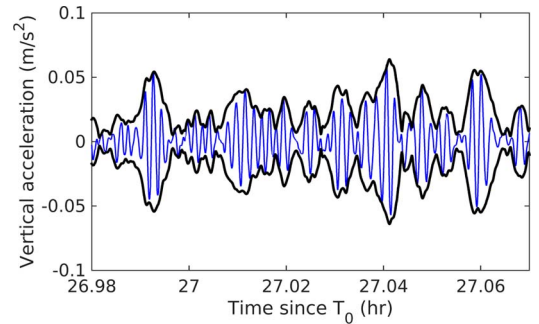


Fig. 10. Hilbert transform applied on low-pass-filtered wave signal (vertical acceleration generated by ice waves measured by V5). The wave packets are clearly visible.

normalized cross correlation is computed as a function of the time shift τ between two signals as

$$\rho_{ij}(\tau) = \frac{E[(S_i(t) - \mu_i)(S_j(t + \tau) - \mu_j)]}{\sigma_i \sigma_j} \quad (3)$$

where E is the expectation value, i and j are the two sensors considered, S_i is the signal recorded by sensor i , μ_i is its mean value, and σ_i is its standard deviation. The coefficient $\rho_{ij}(\tau)$ then falls between 0 (no correlation) and 1 (signals fully correlated). The wave vertical acceleration and the associated envelope shape obtained from sensors V5, V2, and V7 are used to compute cross correlations during the periods when incoming waves are detected. The vertical acceleration is also integrated twice in time using a first-order scheme to obtain

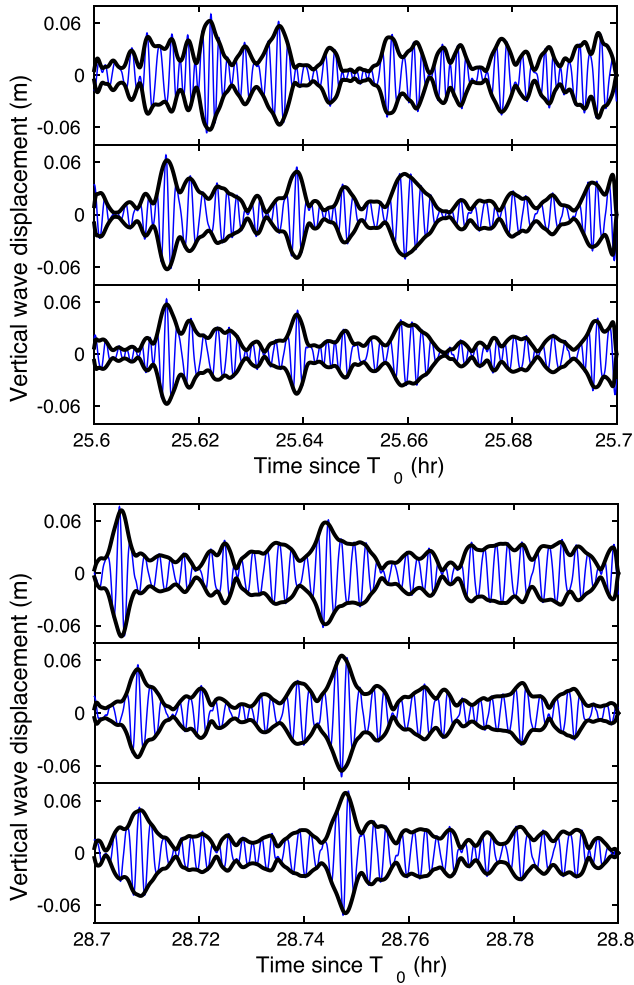


Fig. 11. Comparison of the wave vertical displacement from sensors V5, V2, and V7 (from top to bottom of each figure) before (top) and after (bottom) the transition in incoming-wave frequency distribution. The distance from V5 to V2 is 56 m. The distance between V2 and V7 is 6 m.

the wave amplitude. The resulting elevation contains some drift inherent to the double integration of acceleration data. A high-pass filter of stopband edge 0.07 Hz, passband edge 0.09 Hz, and stopband attenuation 80 dB is generated by the equiripple algorithm and used to correct for the associated drift. The results obtained for wave elevation are presented in Fig. 11.

The cross correlation between the vertical acceleration signals from V2 and V7 and their envelopes computed by Hilbert transform are on the order of 0.8, which indicates that both the individual waves and the packets can be traced between the two sensors. The time shift between the signals is identified by finding the time interval τ for which the cross-correlation function is maximum, using a second-order polynomial fit of the corresponding extremum to get an accuracy level better than the sampling time. The results are shown in Fig. 12. A clear increase can be observed around $T_0 + 27$ h when the high-frequency wind wave component decreases, and the curves for time shifts between the waves and the envelope cross at that time.

Initially, low correlation, starting at about 0.3, is found between sensors V5 and V7. After $T_0 + 27$ h, the correlation increases, i.e., up to about 0.7 at the end of the time series. The surprisingly low correlation in the early time series between

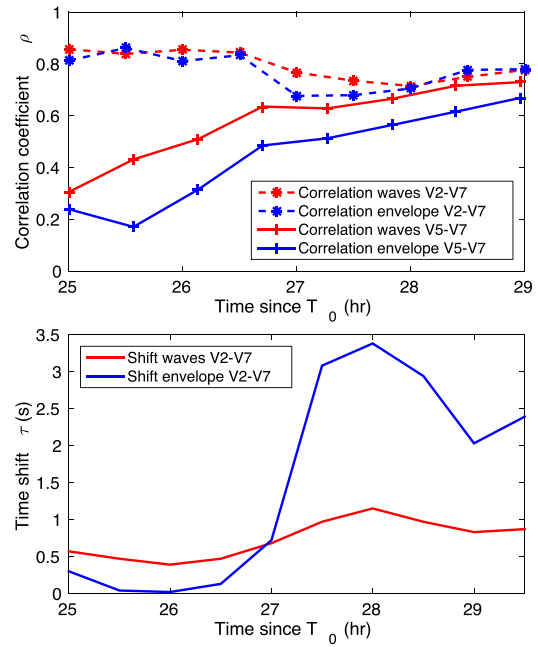


Fig. 12. (Top) Value of the correlation coefficient obtained from vertical wave acceleration of sensors V2, V5, and V7. (Bottom) Time shift τ between the wave and envelope signals from accelerometers V2 and V7.

IMU signals separated by only around 60 m is visible in Fig. 11. While the similarities between the envelopes of signals V2 and V7 are clearly identifiable, the envelope of the signal from sensor V5 looks different before about $T_0 + 27$ h. By contrast, from $T_0 + 27$ h, the wave packets can be traced between the three IMUs. Transition from relatively high-frequency wind-generated waves to lower-frequency swell may explain the difference in the observed correlation, since higher-frequency waves are expected to be more damped and affected by the ice cover.

The crossing of the correlation time shift curves between the wave signals and their envelopes presented in Fig. 12 can be qualitatively explained from results presented in [23], [45], and [46]. The dispersion relation for waves propagating in solid ice reads

$$\omega^2 = \frac{gk + Bk^5 - Qk^3}{\coth(kH) + kM} \quad (4)$$

in which $H = 80$ m is the water depth, and B , Q , and M describe the effects of bending, compression, and inertia of the ice, respectively. Since the measurements were performed on landfast ice far in a fjord system, at a location where maximal tidal currents do not exceed 0.05 m/s [47], no Doppler correction is needed. The parameters in (4) are given by

$$B = \frac{Eh^3}{12\rho_w(1-\nu^2)} \quad Q = \frac{Ph}{\rho_w} \quad M = \frac{\rho_i h}{\rho_w} \quad (5)$$

where $\omega = 2\pi f$ is the angular frequency, $k = |\vec{k}| = 2\pi/\lambda$ is the wavenumber, $g = 9.81$ m/s² is the acceleration of gravity, and $h = 0.8$ m is the ice thickness that was measured at the location of the IMUs. We use standard values for the ice properties that were not measured in the field [48]: $E = 2 \times 10^9$ N/m² for the elastic modulus, $\nu = 0.3$ for the Poisson coefficient,

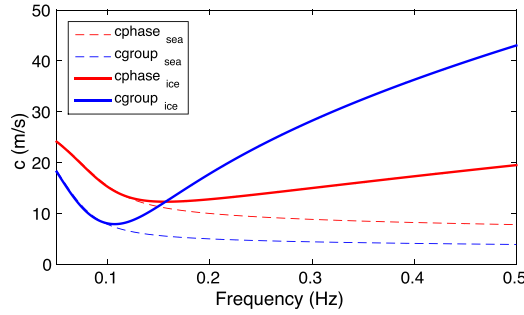


Fig. 13. Phase and group velocity from the dispersion relation (4), corresponding to waves propagating in water covered by 80 cm of ice. Dashed line corresponds to the results for linear waves in the deep-water approximation.

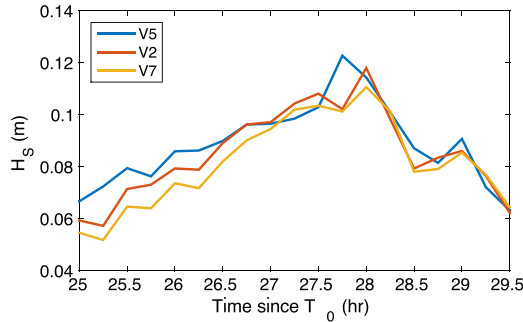


Fig. 14. Comparison of the significant wave height between sensors V5, V2, and V7.

$\rho_w = 1025 \text{ kg/m}^3$ for the volumetric mass density of sea water, and $\rho_i = 920 \text{ kg/m}^3$ for the volumetric mass density of sea ice. Several works have investigated the magnitude of the compression stress P in landfast ice [49], [50], and the value they measured on the field implies that the effect coming from the Qk^3 term can be neglected compared with the other terms. The phase velocity $c_\phi = \omega/k$ and group velocity $c_g = \partial\omega/\partial k$ are calculated from the dispersion relation (4) and presented in Fig. 13 together with the ones corresponding to the deep-water wave dispersion. The dispersion relation is close to that without ice for the lower frequencies, but for frequencies higher than 0.15 Hz, the effect of the ice cover becomes predominant. In the frequency range between 0.1 and 0.15 Hz, which includes the waves measured, a transition between the two regimes is observed. The curves for group and phase velocity cross at around 0.15 Hz, and under 0.12 Hz, flexural effects have a negligible effect on the dispersion relation, which explains the crossing of the time shifts around $T_0 + 27 \text{ h}$ (Fig. 12). Before that, the signal is dominated by shorter waves with higher active frequency around 0.17 Hz (see Fig. 8). The group velocity is then higher than the phase velocity, and therefore, the time shift between the sensors is lower for the wave envelopes than for the waves. On the contrary, after $T_0 + 27 \text{ h}$, the peak frequency is around 0.11 Hz (see Fig. 8), and the phase velocity is higher than the group velocity, which means that the time shift observed between the sensors is higher for the wave envelope than for the phase. This is compatible with the analysis of the same data set performed using different mathematical tools that is presented in [26], which suggests that significant flexural effects affect wave propagation in the early part of the wave signal.

The wave vertical acceleration is used to obtain the evolution in time of the significant wave height. Each vertical acceleration data set is cut in signal intervals of 30 min, with a 50% overlap. Wave elevation is obtained from the same method that was used to generate the data presented in Fig. 11. Significant wave height is computed from the corresponding wave elevation using $H_S = 4 \times \text{std}(\eta)$, where std is the standard deviation, and η is the wave elevation previously computed.

Results are presented in Fig. 14. The significant wave height reaches a peak of 12 cm at around $T_0 + 28 \text{ h}$. The time when peak wave height is reached globally agrees with the numerical model data presented in Section III-A. The significant wave height achieved is significantly less than the value predicted outside the entry of the fjord, and the corresponding reduction can be due to the fjord shape, the presence of the ice cover, and reflections at the edge of the ice cover.

Fig. 14 suggests that some attenuation mechanisms take place in the early phase of the signal (up to $T_0 + 27.5 \text{ h}$), before diminishing. The grouping of the curves around the same value of H_S takes place when the shift from dominating high-frequency wind waves to dominating lower-frequency swell is observed and corresponds with the fact that wave damping and scattering by the ice cover is expected to increase with wave frequency. Since only three sensors are used and the distance between them is only, at most, one wavelength, it is not possible to compute a damping coefficient with a meaningful error bar.

V. CONCLUSION

IMUs were assessed in the laboratory and successfully used on the field in Svalbard to perform measurements of waves in landfast ice. Several advantages can be found in this approach. Because the cost of electronics components has considerably reduced over the last two decades and the electronics has dramatically increased in performance, more capable instruments can now be deployed that are able to measure time series for several variables simultaneously. This is of critical importance when considering the scarcity of data sets available and the richness of the associated physics. The IMUs used present another advantage, since they record the full three components of linear and angular acceleration as well as yaw, pitch, and roll. During a deployment of the instruments on landfast ice in Svalbard, a clear wave signal over one order of magnitude above the noise level of the sensors was detected. The use of IMUs allowed to measure ice acceleration in the horizontal as well as vertical directions and to show that horizontal acceleration in the direction of the fjord had a measurable value of comparable order of magnitude as the vertical one. While surprising, this motion is found consistently in the whole data set and is well above the noise level of the IMUs used. Further analysis of the relation between horizontal and vertical acceleration allows incoming and reflected waves to be detected. Reflected waves are found to be more important at higher frequencies. Additional wave characteristics are analyzed, such as the correlation between the signals from different sensors. Complex features can be observed when comparing the signals, and a clear change takes place at around $T_0 + 27 \text{ h}$. It was possible to relate the corresponding changes with the transition from wind-generated waves to a regime where incoming swell dominates and to explain some of the features observed from

the dispersion relation for waves in ice-covered water. The development of cracks in the ice cover can also participate in this transition. Finally, some damping effect could be observed, but the short distance between the sensors did not allow for accurate estimates of the damping coefficient. Moreover, the observed damping may be the result of a complex interaction of the incoming waves with cracks and regions of variable ice thickness, which are known to produce reflection and scattering effects.

ACKNOWLEDGMENT

The project and field campaign were carried out as a collaboration between the University of Oslo, the University Center in Svalbard, the National University of Ireland, and the Norwegian Meteorological Institute. The authors would like to thank Prof. A. Marchenko, Prof. E. Morozov, Dr. E. K. Lindström, and Laboratory Engineer O. Gundersen for the help during the measurement campaign in Svalbard.

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